

- (1) Show that the elements of F_{p^2} which are fixed by the Frobenius $Fr(x) = x^p$ are exactly the elements of $F_p \subset F_{p^2}$.
- (2) Show that if $z = x + y\sqrt{d} \in F_p[\sqrt{d}]$ where d is not a square mod p , then $z^p = x - y\sqrt{d}$.
- (3) Let $z \in F_p[\sqrt{d}]$ where d is not a square mod p . Define $N(z)$. And show that $N(z) = 0$ iff $z = 0$.
- (4) Show that $N(z_1 \cdot z_2) = N(z_1) \cdot N(z_2)$.
- (5) Let $z \in F_p[\sqrt{d}]$ where d is not a square mod p . How do you compute z^{-1} ? Eg $(2 + 3\sqrt{3})^{-1}$ in $F_{17}[\sqrt{-3}]$.
- (6) Define $T(p)$ in F_{p^2} as the set of elements of norm 1. (You should think of $F_{p^2} = F_p[\sqrt{d}]$ where d is not a square mod p .)
- (7) Describe $T(p)$ in terms of a primitive $p^2 - 1$ root of 1. Which elements of $T(p)$ are squares?
- (8) Show that $T(p)$ is a subgroup of $F_{p^2}^*$ of order $|T(p)| = p + 1$.
- (9) Show that $\beta \in T(p)$ is a square if $\beta^{(p+1)/2} = 1$ (resp. is not a square if $\beta^{(p+1)/2} = -1$).
- (10) Compute $T(7) \subset F_7[i]$ and find the generators of $T(7)$. (Do the same for $T(13) \subset F_{13}[\sqrt{2}]$).

Solutions:

- (1) By FLT every element of F_p satisfies $x^p = x$ and hence is fixed by the Frobenius. Since F_{p^2} is a field, $x^p = x$ has at most p solutions and so F_p is the complete list of all elements fixed by the Frobenius.
- (2) $z^p = (x + y\sqrt{d})^p = x^p + y^p \sqrt{d}^p = x + y\sqrt{d} \cdot d^{(p-1)/2} = x - y\sqrt{d}$. (We have used FLT to conclude $x^p = x$ and $y^p = y$, the fact that in characteristic p we have $(a + b)^p = a^p + b^p$ and the fact that $d^{(p-1)/2} = -1$ since d is not a square mod p).
- (3) If $z = x + y\sqrt{d}$, then $N(z) := x^2 - dy^2$ (you can view this as $z \cdot \bar{z}$ where $\bar{z} = x - \sqrt{d}$.) If $N(z) = 0$, then $x^2 - y^2 d = 0$. If $y = 0$, then $x^2 = 0$ and so $x = 0$. If $y \neq 0$, then $(xy^{-1})^2 = d$ so that d is a square mod p . Impossible.
- (4) We have

$$N(z_1 \cdot z_2) = N((x_1 + y_1\sqrt{d}) \cdot (x_2 + y_2\sqrt{d})) = N(x_1x_2 + y_1y_2d + (x_1y_2 + x_2y_1)\sqrt{d})$$

$$(x_1x_2 + y_1y_2d)^2 - (x_1y_2 + x_2y_1)^2d = x_1^2x_2^2 - dy_1^2x_2^2 - dx_1^2y_2^2 + d^2y_1^2y_2^2 =$$

$$(x_1^2 - dy_1^2)(x_2^2 - dy_2^2) = N(z_1) \cdot N(z_2).$$

- (5) If $z = x + y\sqrt{d}$, then $z^{-1} = \bar{z} \cdot N(z)^{-1} = (x - y\sqrt{d})(x^2 - dy^2)^{-1}$. Check: $z \cdot \bar{z} / N(z) = N(z) / N(z) = 1$. Eg. $(2 + 3\sqrt{3})^{-1} = (2 - 3\sqrt{3})(4 - 27)^{-1} = (2 - 3\sqrt{3})11^{-1} = (2 - 3\sqrt{3}) \cdot (-3) = -6 + 9\sqrt{3}$
- (6) Suppose that $z_1, z_2 \in T(p)$, then $N(z_1) = N(z_2) = 1$ so that $N(z_1 \cdot z_2) = N(z_1) \cdot N(z_2) = 1$ and hence $(z_1 \cdot z_2) \in T(p)$. Similarly if $z \in T(p)$, then $N(z) = 1$ so that $N(\bar{z}) = 1$ and $z \cdot \bar{z} = 1$. But then $z^{-1} = \bar{z} \in T(p)$. Finally $1 \in T(p)$.
- (7) If γ is a primitive $p^2 - 1$ root of 1, then $\alpha = \gamma^{p-1}$ is an element of $T(p)$. To see this, we compute its norm

$$N(\alpha) = \alpha \cdot \bar{\alpha} = \alpha \cdot \alpha^p = \alpha^{p+1} = (\gamma^{p-1})^{p+1} = \gamma^{p^2-1} = 1.$$

It then follows that $T(p) = \{\alpha^k | k = 0, 1, 2, \dots, p\}$. The squares are the elements $\{\alpha^{2k} | k = 0, 1, 2, \dots, (p-1)/2\}$.

(8) Clear from the previous exercise.

(9) We may write $\beta = \alpha^k$ where $\alpha^{p+1} = \gamma^{p^2-1} = 1$. Note that $\alpha^{(p+1)/2} = \gamma^{(p^2-1)/2} = 1$ (it is not 1 and it is a square root of 1).

If $k = 2j$, then $\beta^{(p+1)/2} = (\alpha^{p+1})^j = 1^j = 1$. If $k = 2j + 1$, then $\beta^{(p+1)/2} = (\alpha^{p+1})^j \cdot \alpha^{(p+1)/2} = 1^j \cdot (-1) = -1$.

(10) $T(7) = \{\pm 1, \pm i, \pm(2+2i), \pm(2-2i)\}$ (just check that $N(z) = 1$ for each element, eg. $N(2-2i) = 4 + 43 = 8 = 1$). The generators are $\pm(2+2i), \pm(2-2i)$ (check!).

$T(13) = \{\pm 1, \pm 3 \pm 2\sqrt{2}, \pm 5 \pm 5\sqrt{2}, \pm 4 \pm \sqrt{2}\}$ (the point is that $1 = 1^2 = 3^2 - 2 \cdot 2^2 = 5^2 - 2 \cdot 5^2 = 4^2 - 2$ modulo 13).